

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 9455af9b51a480be

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_16_qam.iq
File SHA-256 Digest : 9455af9b51a480bef89d4558de9b16ab718af8f8c73d5fe5c5b4ffc36e91877e
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:09:36Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+0.689 kHz	Spectral Centroid Offset
Estimated Baud Rate	500.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	1.46 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1734.38 kHz	Cumulative PSD 99% Energy
Estimated SNR	6.39 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	3.67 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1373	Wiener Entropy Index (0.0 – 1.0)

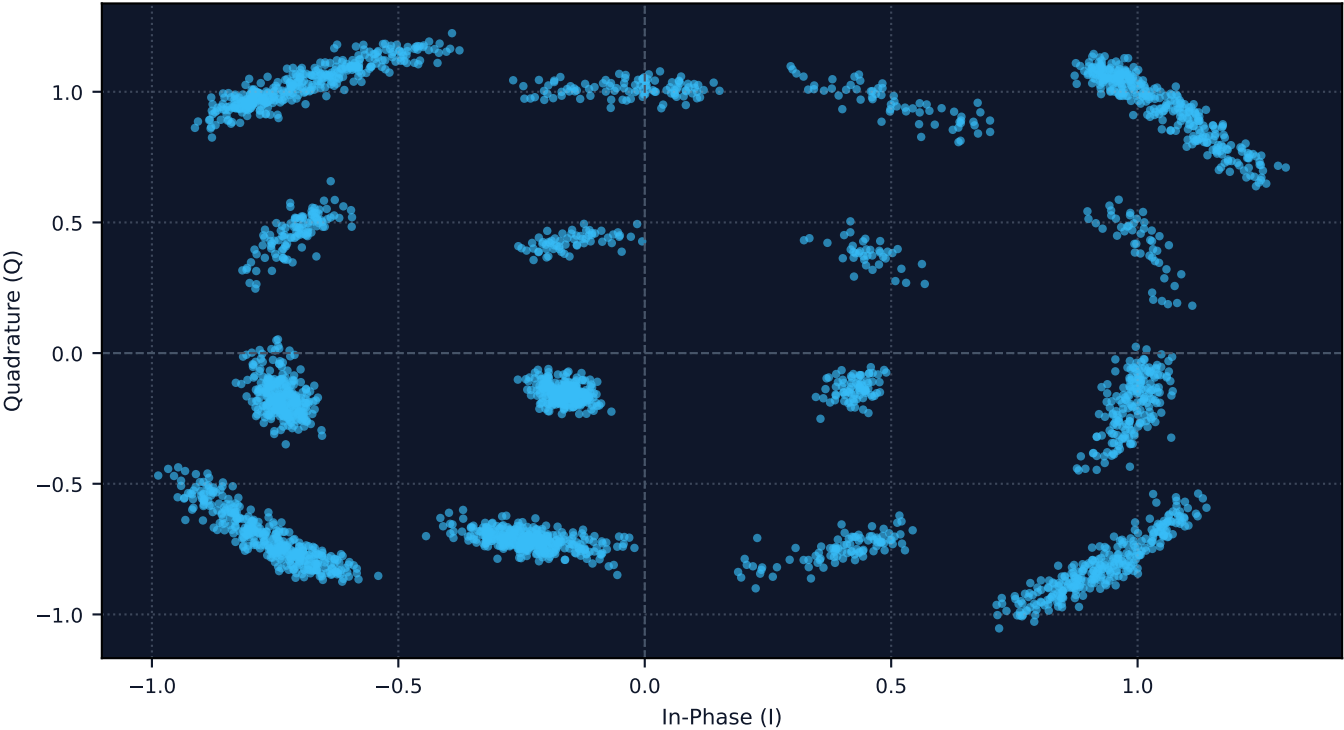
PREDICTED MODULATION : QPSK

CLASSIFIER CONFIDENCE : 70.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 16,384 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [QPSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	589	0x0000	0	1967 B	CRC MISMATCH	PARTIAL_RECOVER
#2	2247	0x0000	1	1760 B	CRC MISMATCH	PARTIAL_RECOVER
#3	3076	0x0000	2	1656 B	CRC MISMATCH	PARTIAL_RECOVER
#4	3905	0x0000	3	1553 B	CRC MISMATCH	PARTIAL_RECOVER
#5	4734	0x0000	4	1449 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 FFD177377DF7117FFBB9FF17DFDFDF7F [FRAME 1 PAYLOAD]